

Curricular Internship Report: Clumping Factor Calculations in Cosmological Simulations

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Curricular Internship Report

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Abstract

This report summarizes the first month of a four-month curricular internship focused on clumping factor calculations in cosmological simulations. The scientific objective of the internship is to measure the clumping factors predicted by simulations with different dark matter models and compare those predictions with observational or experimental constraints, potentially restricting the allowed parameter space of the models. The first part of the work developed the scientific background needed to understand dark matter, cosmological simulations, reionization, the intergalactic medium, and clumping-factor definitions. The later part implemented and tested a reproducible computational pipeline for gas and dark-matter clumping factors, scaled the calculation to larger simulation outputs, and began an extension toward Lyman-alpha forest observables.

I. INTRODUCTION

The scientific objective of the internship is to understand and implement clumping factor calculations for cosmological simulation data. Over the full four-month project, the goal is to compute the clumping factors predicted by simulations with different dark matter models and compare those predictions with observational or experimental constraints. If the predicted clumping factors differ in a measurable way between models, this comparison could help restrict the allowed model parameters.

In the context of reionization, the clumping factor measures how inhomogeneous matter is relative to a uniform distribution. Since recombination rates scale approximately with density squared, small-scale density structure can strongly affect the ionization state of the intergalactic medium (IGM). A reliable clumping factor calculation therefore needs both physical care, because gas and dark matter trace structure differently, and computational care, because modern simulations are large enough that naive full-snapshot processing can become impractical.

The work described here had two connected goals. The first was scientific: to build the background needed to understand what is being measured, why different dark matter models may change the relevant density structure, how cosmological simulations store particle and gas data, and how clumping factors enter reionization calculations. The second was practical: to develop the initial implementation into a reproducible analysis workflow with command-line tools, tests, cluster submission scripts, benchmark outputs, and documentation.

II. SCIENTIFIC BACKGROUND

The first two weeks and part of the third week were primarily devoted to background research. This stage covered the physical role of dark matter in structure formation, how cosmological simulations represent matter fields, the basic history and physics of cosmic reionization, and the reason clumping factors appear in recombination-rate estimates. A central lesson from this phase was that a clumping factor is not just a software diagnostic: it depends on the matter component being measured, the density field used to define the IGM, the smoothing or gridding method, and the averaging convention.

Modern cosmological simulations evolve matter from early-Universe initial conditions into the nonlinear structures observed at later times. They are especially useful because they connect assumptions about cosmology and dark matter to the distribution of galaxies, halos, gas, and the IGM [1]. In this project, the relevant point is that changing the dark matter model can modify small-scale structure formation. Those changes can alter the abundance of small halos, the density field around collapsed structures, and the distribution of gas that participates in recombination and absorption.

In practice, the simulation fields are not continuous. Dark matter is represented by simulation particles, each of which stands for a large mass element rather than an individual physical particle. To estimate a density field from these particles, their masses must be assigned to a mesh or smoothed over a characteristic scale, typically related to the grid spacing or the local interparticle distance. The gas is represented differently in moving-mesh simulations: it is defined on cells of a tessellation, so each gas element has a volume, density, position, and thermodynamic state. This difference matters for clumping calculations, because the dark matter calculation begins from discrete massive tracers, while the gas calculation begins from finite-volume cells whose native volumes are already part of the simulation output.

A commonly used density clumping factor has the form

$$C = \frac{\langle \rho^2 \rangle}{\langle \rho \rangle^2}. \quad (1)$$

If the density field is uniform, $C = 1$. If the field contains dense structures, $C > 1$. For reionization, this is important because dense regions recombine more efficiently and therefore increase the number of ionizing photons required to keep the Universe ionized. Observational

work has also connected the clumping factor to quantities such as the ionizing mean free path and photoionization rate near the end of reionization [2]. This connection is important for the long-term goal of comparing simulation predictions with observational constraints.

However, the clumping factor is not uniquely defined. It depends on the averaging volume, the gas included in the IGM, and whether the calculation uses a hard overdensity threshold or a more physically motivated selection. Recent work on recombination clumping factors shows that changing the IGM definition can significantly change the inferred value of C [3]. This motivates treating the definition of the clumping factor as part of the scientific model, not only as an implementation detail.

The Lyman-alpha forest is also relevant to this project because it gives observational access to the neutral hydrogen distribution through absorption along lines of sight to distant sources. Instead of directly observing the full three-dimensional density field, one observes transmitted flux and absorption statistics. We plan to use Lyman-alpha forest calculations as a bridge between simulation fields and observations: the goal is to determine whether an observationally inferred clumping factor can be defined from Lyman-alpha quantities and how that quantity relates to the more standard density-based clumping factor computed directly from the simulations. Modeling the Lyman-alpha forest requires care because the absorption depends not only on density, but also on gas temperature, ionization state, velocities, and the relation between baryons and the underlying matter distribution [4].

III. INITIAL IMPLEMENTATION

The first coding stage consisted of standalone scripts for gas and dark-matter clumping-factor calculations. These scripts separated gas and dark-matter calculations and explored direct gas calculations, dark-matter calculations, spherical and cubic smoothing strategies, and Pylians-based grid calculations for comparison with custom implementations.

This initial implementation was important because it produced the first working results and clarified the computational structure of the problem. It also showed which parts of the workflow had to become more systematic: the same physical definition had to be applied consistently across gas and dark matter, results had to be saved in a comparable format, and the same calculation had to be repeatable across different simulations, resolutions, and numerical methods.

IV. DEVELOPMENT OF A REPRODUCIBLE COMPUTATIONAL PIPELINE

After the initial implementation, the work was organized into the `clumping_factor` Python package, with command-line entry points for computing and plotting results. This separated the expensive scientific calculation from later plotting and made result files the stable interface between compute jobs and local analysis.

The implementation writes JSON summaries and supports several numerical methods:

- `sphere`, using spherical top-hat smoothing;
- `cube`, using cubic smoothing;
- `pylians`, using Pylians mass assignment and smoothing where available;
- a volume-weighted native gas-cell method, which uses the simulation cell volumes directly instead of first depositing the gas onto a uniform grid;
- an optical-length-weighted IGM method under development, which uses neutral hydrogen information to weight gas by its effective optical length rather than applying only a hard overdensity threshold.

The core density clumping calculation can be summarized as

$$C_\rho(\Delta_{\text{th}}) = \frac{\langle \rho_{\text{target}}^2 \rangle_{\text{selected}}}{\langle \rho_{\text{target}} \rangle_{\text{selected}}^2}, \quad (2)$$

where selected cells are chosen using an overdensity threshold. The package also supports cases where the field used to define the mask differs from the field used to measure the target clumping factor. This is useful, for example, when selecting IGM cells from one matter field and measuring gas clumping inside that selected region.

The validation work was designed to make the calculation reliable enough for comparisons between dark matter models. Instead of only checking that a script ran to completion, the tests verify that command-line options are interpreted consistently, grids are constructed as expected, chunked loading agrees with the intended data flow, native gas-cell calculations preserve the correct volume-weighted averaging convention, and plotting operates on the saved result files. This gives confidence that later differences between simulations are more likely to reflect physics than accidental differences in how a run was configured.

V. PERFORMANCE, SCALABILITY, AND CLUSTER WORKFLOW

After the package structure was established, a major focus was scaling the calculations to larger simulation outputs. The repository history between June 8 and June 15 shows the addition of chunked loading, logging, parallelized calculations, same-node worker pools, timing diagnostics, batch processing, and new job submission scripts.

The chunked workflow avoids loading full snapshots into memory when this would be unsafe. For gridded calculations, snapshot data are streamed in chunks and deposited onto density grids. Same-node parallelism then partitions snapshot files or file ranges across workers. Each worker builds private grid accumulators, writes temporary grid outputs, and the parent process reduces those outputs into final grids. This design reduces memory pressure and makes it possible to run calculations on cluster nodes with explicit memory limits.

The cluster workflow was also formalized through PBS submission scripts. These scripts encode choices such as grid size, numerical method, particle type, thread count, memory limits, wall time, and throttled job submission. Throttling is important because multiple simultaneous jobs reading the same large simulation output can create filesystem contention. The result JSON files include timing diagnostics such as grid-build time, worker stream time, reduction time, HDF5 I/O time, and the slowest worker runtime. These diagnostics make the benchmarks interpretable rather than just collections of final numbers.

VI. RESULTS AND ANALYSIS

The second half of the month produced several result batches and analysis figures. These include TNG100-3 comparisons, THESAN scaling runs, Thesan-1 and Thesan-2 comparisons, numerical-method comparisons, and dark-matter Pylans grid/thread scaling experiments. The analysis products make it possible to compare both scientific outputs, such as clumping consistency across methods, and computational behavior, such as wall time and scaling with thread count.

Figure 1 shows an example performance dashboard for the Thesan-1 versus Thesan-2 scaling analysis. This type of plot is useful because it connects the scientific calculation to the practical question of which numerical method, grid size, and parallel configuration can

be run efficiently.

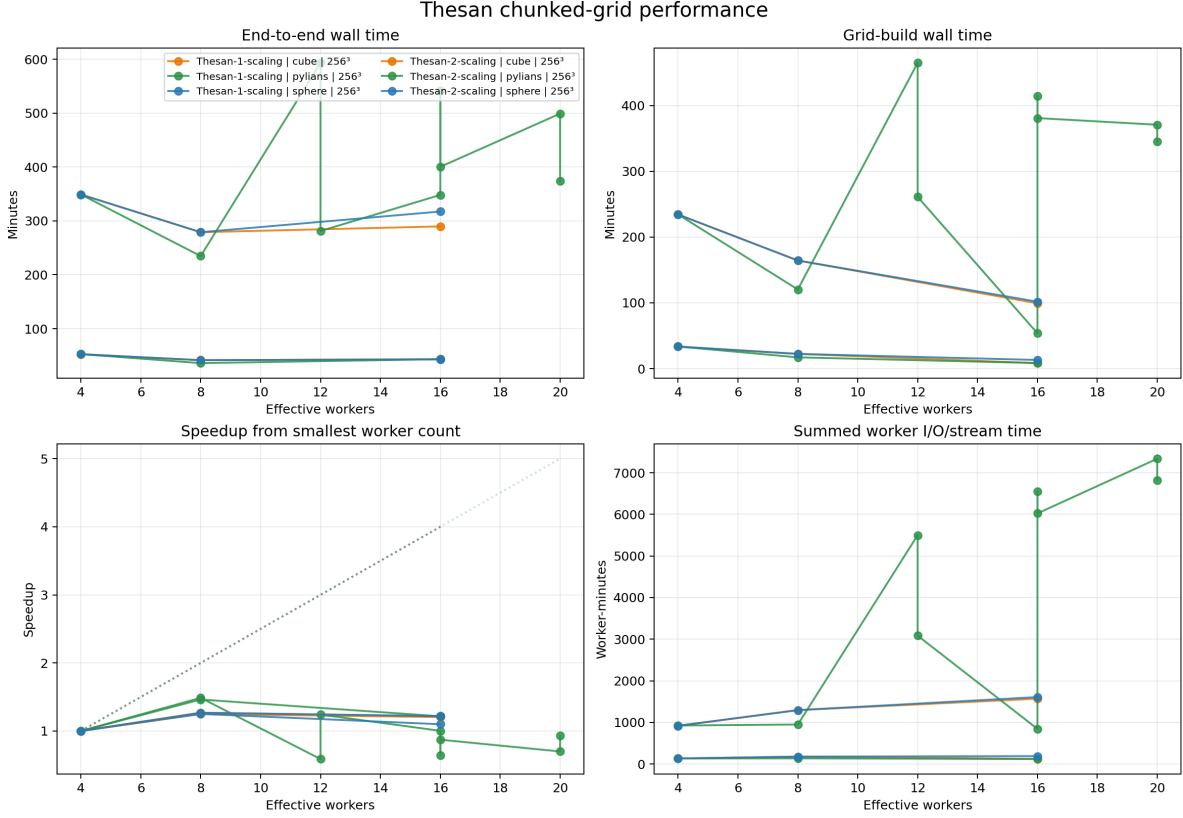


FIG. 1. Performance dashboard for Thesan-1 versus Thesan-2 scaling results. The figure summarizes timing behavior across benchmark runs and supports comparison of parallel configurations.

Figure 2 summarizes clumping consistency for the same comparison. This is the scientific counterpart to the timing analysis: the goal is not only to run faster, but to understand whether different computational configurations preserve the same physical result.

Additional analysis focused on dark-matter Pylians scaling across grid sizes and thread counts. This work is useful for separating physical differences from numerical and computational effects. Figure 3 shows a grid-scaling diagnostic from these runs.

VII. MILESTONES

Table I groups the month of work into phases. The table is intentionally thematic rather than a chronological commit log, because the main outcome of the internship month was a coherent scientific and computational workflow.

Period	Main activity	Outcome
Weeks 1–2 and part of week 3	Background research	Built conceptual foundation in dark matter, simulations, reionization, IGM structure, and clumping-factor definitions.
Initial implementation	Standalone gas and dark-matter scripts	Produced first working clumping-factor calculations and comparison outputs using separate prototype scripts.
June 4–5	Pipeline structure and early validation	Introduced CLI package structure, environment setup, tests, plotting style updates, overdensity standardization, and comparisons with the initial implementation.
June 8–13	Chunked and parallel computation	Added chunked loading, logging, parallel grid construction, timing diagnostics, batch processing, and improved cluster scripts.
June 15–19	THESAN and benchmark analysis	Uploaded and analyzed result batches, added new parallelization techniques, compared Thesan-1 and Thesan-2, and improved analysis plots.
June 22–23	New physics direction	Added Lyman-alpha forest and optical-length-weighted IGM groundwork, safeguards, and new results for analysis.

TABLE I. Summary of the first month of internship work, grouped by conceptual and technical phase.

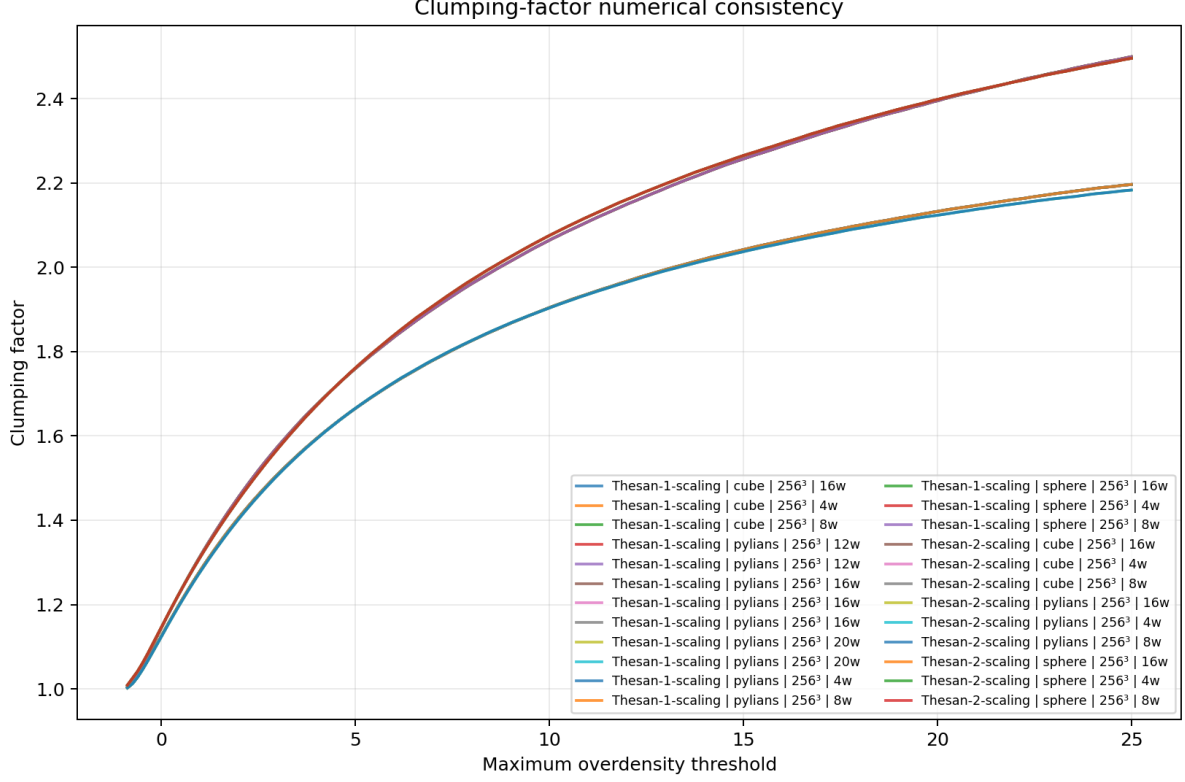


FIG. 2. Clumping consistency plot for the Thesan-1 versus Thesan-2 comparison. This figure is used to check whether method and simulation choices produce consistent clumping-factor behavior.

VIII. CURRENT WORK AND FUTURE DIRECTION

The newest direction is motivated by the need to relate the standard density-based calculation to a quantity that can be inferred more directly from absorption. The current density-only calculation measures moments of mass density under an overdensity selection, while an optical-length-weighted IGM calculation requires ionization state, electron density, recombination physics, neutral-hydrogen gradients, and a soft weight describing how much each region contributes to absorption. In simplified form, the target quantity discussed in the comparison notes has the structure

$$C_{\text{rec},F} = \frac{\langle \alpha(T) n_{\text{HII}} n_e F \rangle_V}{\langle \alpha(T) \rangle_V \langle n_{\text{HII}} \rangle_V \langle n_e \rangle_V \langle F \rangle_V}, \quad (3)$$

where $F = \exp(-\tau_{\text{eff}})$ acts as a soft weight related to the effective optical length of the gas. This is physically distinct from simply applying a density threshold or weighting a density variance.

Current uncommitted work should be understood as active development rather than com-

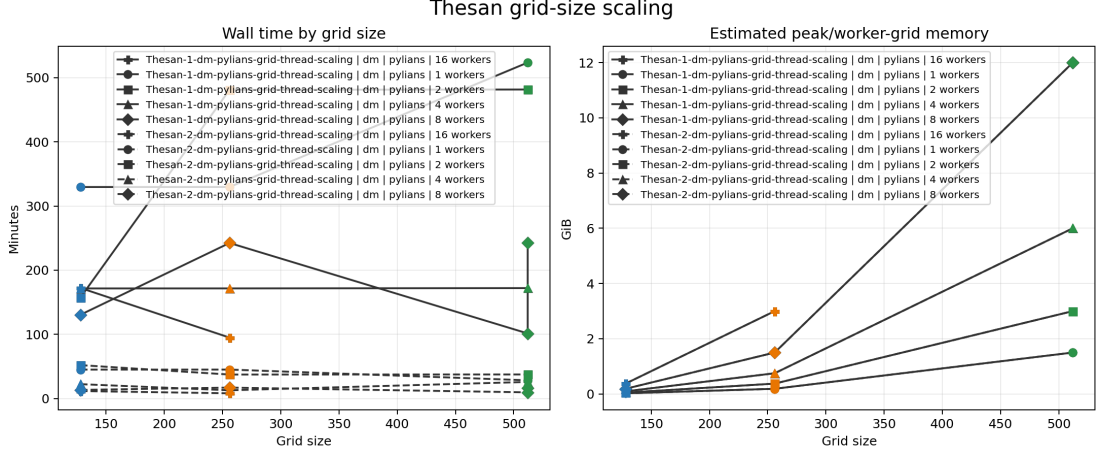


FIG. 3. Dark-matter Pylans grid-scaling diagnostic. These benchmarks help identify how resolution and thread count affect both performance and numerical behavior.

pleted history. It includes changes to the README, job scripts, result handling, plotting, CLI options, and tests. The purpose of this work is to continue connecting the density-based calculation to the newer Lyman-alpha and optical-length-weighted IGM directions.

The next steps are:

- validate the optical-length-weighted IGM calculation against expected THESAN gas fields;
- extend redshift-dependent calculations across more snapshots;
- compute clumping factors for simulations with different dark matter models;
- compare standard density-based clumping factors with Lyman-alpha-inferred quantities;
- determine whether the resulting differences can constrain dark matter parameters.

IX. CONCLUSION

The first month of the internship produced both scientific understanding and a practical computational workflow. The early research phase established why clumping factors matter for reionization and why the definition of the density field, smoothing method, and

IGM selection all affect the result. The initial scripts then provided the first working implementation. The later development transformed that prototype into a tested package with command-line tools, cluster job scripts, chunked and parallel execution, benchmark diagnostics, result analysis, and documentation.

The project is now positioned to move from density-based clumping-factor calculations toward more physically specific recombination and optical-length-weighted measures. This progression reflects the main arc of the month: learning the science, prototyping the calculation, making it reproducible, scaling it to larger simulations, analyzing the outputs, and identifying the next scientific extension.

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